

TESTING SQL

Higher Computing Science

TESTING SQL

- SQL queries should be tested by:
 1. producing a table of expected results
 2. running the queries and checking that the actual results match the expected results

EXAMPLE 1

Extracts from a car hire database are shown below:

Customer								
customerID	surname	telephone	postcode					
C001	Green	07554137612	EH12 7TT					
...	...	...	...					
C012	Grimes	0777543012	Booking					
				bookingRef	customerID	registration	daysBooked	startDate
				B3481	C005	BA18 HER	3	17/04/2020
				...	...	...	...	

EXAMPLE 1

A query is required to show details of car hire bookings where the number of days booked is more than 5.

The expected output is shown below:

Customer Surname	Customer Telephone	Number of Days Booked
Richards	07538124325	7
Watson	07885332663	7
MacArthur	07664561821	7
Grimes	07775430192	6

EXAMPLE 1

A query is required to show details of car hire bookings where the number of days booked is more than 5.

The SQL query used is shown below, but does not work as expected. Why?

```
SELECT surname, telephone, daysBooked  
FROM customer, Booking  
WHERE daysBooked > 5;
```

EXAMPLE 1

A query is required to show details of car hire bookings where the number of days booked is more than 5.

The SQL query used is shown below, but does not work as expected. Why?

1. An alias should have been used to display the readable column headings
2. There should be a JOIN:

```
AND customer.customerID = booking.customerID
```

EXAMPLE 1

A query is required to show details of car hire bookings where the number of days booked is more than 5.

Correct SQL

```
SELECT surname AS [Customer Surname], telephone AS  
[Customer Telephone], daysBooked AS [Number of Days  
Booked]  
  
FROM customer, Booking  
  
WHERE daysBooked > 5  
  
AND customer.customerID = booking.customerID;
```

EXAMPLE 2

Extracts from a car hire database are shown below:

Car					
registration	make	model	colour	depotID	mileage
BA18 HER	Volkswagen	Golf	Black	D1004	21233
...	...	...	...	...	...
SN16 CFR	Nissan	Qashqai	Silver	D1458	38687

EXAMPLE 2

A query is required to show each make of car, together with the highest mileage for a car of that make.

The expected output is shown below:

make	Highest Mileage for Make
Ford	102300
Mazda	84662
Mercedes	30043
Nissan	61752
Vauxhall	32100
Volkswagen	45208

EXAMPLE 2

A query is required to show each make of car, together with the highest mileage for a car of that make.

The SQL query used is shown below, but does not work as expected. Why?

```
SELECT make, COUNT(mileage)
FROM car
ORDER by make ASC;
```

EXAMPLE 2

A query is required to show each make of car, together with the highest mileage for a car of that make.

The SQL query used is shown below, but does not work as expected. Why?

1. An alias should have been used to display the readable column heading 'Highest Mileage for Make'
2. The `COUNT ()` function has been used instead of `MAX ()`
3. There needs to be a `GROUP BY` statement:
`GROUP BY make`

EXAMPLE 2

A query is required to show each make of car, together with the highest mileage for a car of that make.

Correct SQL

```
SELECT make, MAX(mileage) AS [Highest Mileage for Make]  
FROM car  
GROUP BY make  
ORDER by make ASC;
```